Targeting NELFE Splicing with Antisense Oligonucleotides in Hepatocellular Carcinoma

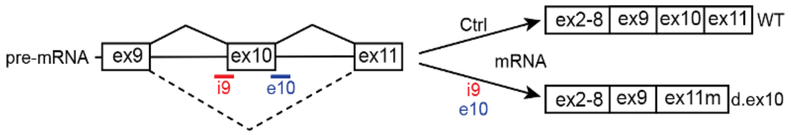
Annabelle Yao[1], Laura Reynolds[2], Hien Dang[2]

[1]The Lawrenceville School, Lawrence Township, NJ, [2]Department of Surgery, Thomas Jefferson University, Philadelphia, PA

Background

- ***Liver Cancer*** is the 6th most common cancer, but has the 3rd highest mortality rate with a 5-year survival rate of 21%[1]
- Typical Causes: Cirrhosis (scarring of the liver, which can also be caused by chronic hepatitis virus and alcohol use)
- ***Hepatocellular Carcinoma (HCC)*** - The most common type of liver cancer[2]
- Starts in cells called hepatocytes (predominant cell type in liver)
  - Responsible for carbohydrate, lipid and protein metabolism, detoxification and immune cell activation to maintain liver homeostasis
- ***Negative Elongation Factor E (NELFE)*** – Regulates Gene transcription by pausing RNA Polymerase II, MYC Signaling promoting HCC progression
  - 1 of top 3 dysregulated differentially expressed RNA binding proteins in HCC + is upregulated[3]
- ***Antisense Oligonucleotides (ASOs)*** – Short synthetic, single stranded oligodeoxynucleotides that alter RNA + modify protein expression by altering splicing[4]
  - ***ASOs targeting human NELFE intron 9- exon 10, exon 10-intron 10 splice junctions (figure 1)***

Figure 1



Goal: to analyse how ASOs affect NELFE isoform production, mRNA stability, and protein levels, potentially offering new therapeutic angles for HCC

Hypothesis: Turning a natural genetic vulnerability (HCC SNP) into a precision treatment tool via ASO-induced exon skipping, can be harnessed therapeutically and administered as microbubble ASOs to reduce NELFE levels and potentially slow cancer progression

Methodology

Figure 2

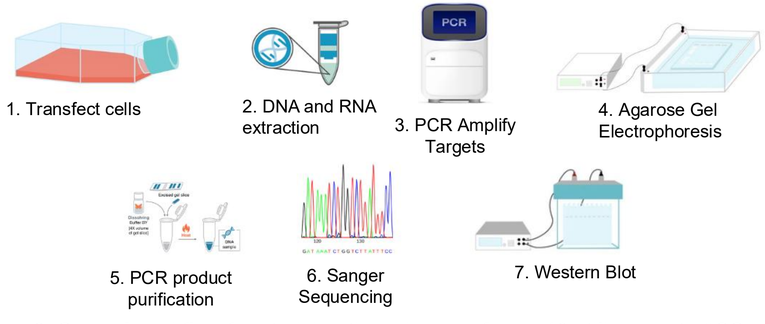


Figure 2: General workflow for studying the skipping of exon 10 utilizing both ASOs and CRISPR-Cas9 system.[5]

Results

Figure 3

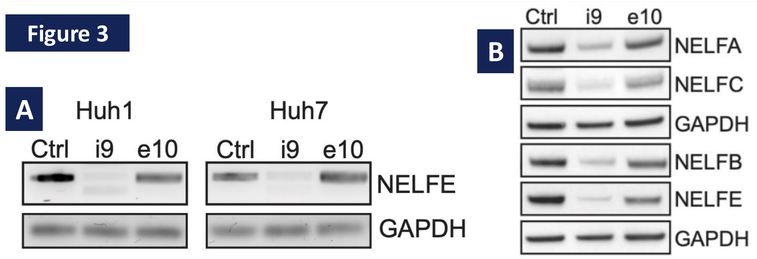


Figure 3: i9 and e10 ASOs alter NELFE splicing, decreasing NELFE expression and impacting broader NELF complex. I9 has larger effect on exon 10 skipping—*A:* both Huh1 and Huh7 cell lines show isoform switching (D.ex-10). ASOs are effectively modulating splicing of the NELFE transcript inducing exon 10 skipping. ***B:*** ASOs reduced NELFE protein in both i9 and e10 conditions. Decreased NELFB and NELFC expression, suggests destabilization of NELF complex.

Figure 4

***Actinomycin D:** A transcriptional inhibitor that binds to DNA at the start of transcription initiation complex, preventing RNA polymerase II from elongating new transcripts

Goal: see if the D.ex-10 isoform degrades faster than the WT NELFE mRNA to understand if the deleted isoform lowers RNA overall and lowers proteins to cause reduced NELFE expression

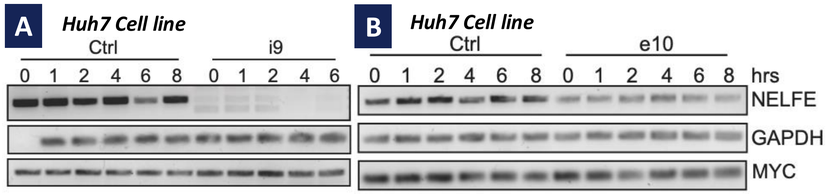


Figure 4: Degradation of D.ex-10 isoform in Huh7 should be faster than WT NELFE mRNA. E10 seems more stable than i9 (i9 degradation happens faster than e10) —*A:* Control gets somewhat fainter as hours increase, change in i9 FL(top band) and D.ex-10(lower band) density over time seems consistent between both. ***B:*** e10 does not show decreases in band strength compared the ctrl over time. If Actinomycin D works, expect MYC to be gone by 2-4 hours. Gapdh: loading control and stable mRNA control; MYC: short half life control.

Figure 5

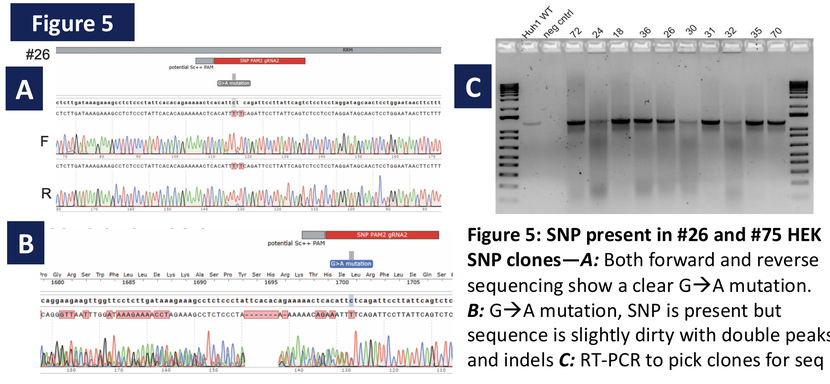


Figure 5: SNP present in #26 and #75 HEK SNP clones—*A:* Both forward and reverse sequencing show a clear G→A mutation. ***B:*** G→A mutation, SNP is present but sequence is slightly dirty with double peaks and indels ***C:*** RT-PCR to pick clones for seq

Conclusions

- Increased NELFE Expression correlates with HCC Disease progression
- SNP HCC patients have less mRNA NELFE expression in tumors than WT patients
- ASOs targeting exon 10 splice junctions of NELFE mimic a naturally occurring SNP and cause alternative splicing of NELFE, impacting the NELF complex
- ASOs i9 and e10 drive exon 10 skipping and result in decreased WT NELFE protein expression
- Lower NELFE disrupts the NELF complex, potentially impairing transcriptional pausing in cancer cells
- I9 has shorter half life than e10 ASO, i9 has faster degradation than control or e10

Future Directions

- Deliver different ASOs to liver tumors *in vivo* using microbubbles, and experiment with different Mechanical Index ultrasound waves
- Assess ASO effectiveness in syngeneic mouse model *in vivo*
- Optimize Actinomycin D mRNA decay assay and validate potential HEK SNP clones

Acknowledgements

Funding and support provided by Thomas Jefferson University Department of Surgery, Sidney Kimmel Cancer Center, and the Lawrenceville School Hutchins Science Scholar Program. AY was also guided by Laura Reynolds and Hien Dang in the lab.

References

[1]Sung, H., Ferlay, J., Siegel, R. L., Laversanne, M., Soerjomataram, I., Jemal, A., & Bray, F. (2021). Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA: A Cancer Journal for Clinicians*, *71*(3), 209–249.

[2]American Cancer Society.*Cancer Facts & Figures 2023*. Atlanta, Ga: American Cancer Society; 2023

[3]Dang, H., Takai, A., Forgues, M., Pomyen, Y., Mou, H., Xue, W., Ray, D., Ha, K. C. H., Morris, Q. D., Hughes, T. R., & Wang, X. W. (2017). Oncogenic Activation of the RNA Binding Protein NELFE and MYC Signaling in Hepatocellular Carcinoma. *Cancer Cell*, *32*(1), 101-114.e8.

[4]Scoles, D.R., E.V. Minikel, and S.M. Pulst, *Antisense oligonucleotides.* A primer, 2019. 5(2): p. e323.

[5]Nguyen, Trish. Targeting NELFE Splicing with Antisense Oligonucleotides and the CRISPR-Cas9 System in Hepatocellular Carcinoma.